DIRECT MODEL OUTPUT · GPU ACCELERATED

AssessClear AI Pre-Pilot Master Synthesis Report

Generated directly by Qwen 3 (14B) on NVIDIA GeForce RTX 2070 SUPER GPU

MODEL ENGINE	COMPUTE ARCHITECTURE	DISTRICTS EVALUATED	STUDENTS ASSESSED
Qwen 3 (14B - Q4_K_M)	NVIDIA RTX 2070 SUPER (CUDA)	Katni, Indore, Bhopal, Raisen	~351 (Grades 6–8, Math & Hindi)

1. Executive Summary

AssessClear AI, a digital assessment tool, was piloted across 9 schools in 4 districts (Katni, Indore, Bhopal, Raisen) to evaluate its efficacy in automating answer-sheet analysis, providing personalized feedback, and supporting educators. While the tool demonstrated **strong potential for remedial learning and multi-tiered feedback**, critical technical and usability challenges emerged, including **OCR inaccuracies, dashboard limitations, network dependency, and inconsistent AI judgments**. This report synthesizes findings, empirical metrics, and recommendations for improvement.

2. Key Findings

2.1 Dashboard & Session Management Issues

- **Evaluation Count Mismatch:** The dashboard counted **questions evaluated** (e.g., 29 evaluations for 3 answer sheets), not **answer sheets assessed**.
- **Data Retention Limit:** A 200-evaluation cap caused older data to disappear, with **23.4% data loss** overall (59.3% in Katni).
- **Session Code Retrieval:** Users struggled to retrieve session codes for subsequent assessments, requiring **manual note-taking**.
- **Dashboard Usability:** Hindi subject and higher grades (7–8) were missing in Indore; history questions were inconsistent.

2.2 Technical Challenges

- **OCR & Image Recognition Errors:**
  - **Digit Recognition Failures:** 5.1% error rate (e.g., `9996 → 3226`).
  - **MCQ Misidentification:** AI misread Option D as B; unattempted questions were sometimes marked as correct.
  - **Blurry Image Handling:** AI attempted to interpret unclear images instead of flagging them.
- **Question Detection Failures:** 10.7% of questions skipped (e.g., Q1/Q2 on first page in Indore/Katni).
- **Network Dependency:** In Raisen (MS Kharbai), **10% zero-network failure** stalled scanning and link opening.

2.3 Feedback Quality & Pedagogical Alignment

- **Strengths:**
  - **Multi-tiered Feedback:** "*What you did well,*" "*Think about this,*" "*Next steps*" were praised for remedial learning.
  - **Hindi Language Capability:** Strong for MP context, though **language accuracy needed improvement**.

- **Weaknesses:**
  - **AI vs. Teacher Judgment Discrepancies:** AI marked a student's "3 months" as incorrect for a question expecting "90 days."
  - **Actionability Gap:** Feedback often focused on **"what to improve"** rather than **"how to do it"** (e.g., "practice antonyms" without classroom activity suggestions).
  - **Inconsistent Feedback Across Pages:** Teacher vs. student feedback in PDFs diverged.

2.4 Time & Resource Burden

- **Processing Time:**
  - AI processing: **38.9s/sheet** (Indore: 22.8s, Bhopal: 73.4s).
  - Human review: **144.5s/sheet** (2.4 min/paper).
  - **Total Mean Time:** 183.4s/sheet (3.06 min).
- **Class-Level Time Burden:**
  - 40-student class: **122.3 min** (2.04 hrs) without summary UI; **23.3 min** with summary UI (81% time saved).

3. Empirical Metrics (TabFM Data)

Metric	Value
Total Students Assessed	351 (Bhopal: 125, Indore: 100, Katni: 91, Raisen: 35)
AI Processing Time	**38.9s/sheet** (Indore: 22.8s, Bhopal: 73.4s, Katni: 26.6s)
Human Review Time	**144.5s/sheet** (2.4 min/paper)
Total Time per Sheet	**183.4s** (3.06 min)
Question Detection Fidelity	**89.6%** (Q1/Q2 skipped in Indore/Katni)
Digit OCR Error Rate	**5.1%** (e.g., `9996 → 3226`)
200-Cap Data Loss Rate	**23.4%** overall (59.3% in Katni)
Zero-Network Failure Rate	**10.0%** (MS Kharbai, Raisen)
Classification Risk Triage	**Critical Blocker:** 10.0% (N=35), **High Risk:** 39.3% (N=138)

4. User Feedback & Recommendations

4.1 From Educators & Observers

- **Positive Aspects:**
  - **Remedial Learning Potential:** AI's step-by-step feedback for incorrect answers was praised.
  - **Offline Support Requested:** 100% of districts highlighted network dependency as a critical barrier.
- **Critical Concerns:**
  - **Teacher Override Needs:** AI should be a **support tool**, not a replacement (Indore, Bhopal).
  - **Competency-Based Dashboards:** Users requested drill-down by student/competency/question.
  - **Confidence Scores:** Needed to prioritize high-risk errors.

4.2 Recommendations

1. **Enhance OCR & Image Handling:**
   - Improve digit recognition (e.g., `9996 → 3226`) and blur detection.
   - Flag unclear images instead of attempting interpretation.
2. **Dashboard Improvements:**
   - Fix evaluation count mismatch and data retention limits.
   - Add Hindi language accuracy checks and support for higher grades (7–8).
3. **Offline & Low-Network Support:**
   - Enable sync for 30–40+ students and open-ended questions.
   - Reduce dependency on stable internet (critical in Raisen).
4. **Teacher Override & Customization:**
   - Allow teachers to review/edit AI-generated feedback.
   - Add confidence scores for high-risk errors.

5. **Time Efficiency:**
    - Implement class-level summary dashboards to reduce review time by 81% (from 122.3 min to 23.3 min for 40 students).

5. Conclusion

AssessClear AI shows **strong potential** for transforming assessment workflows with its **multi-tiered feedback** and **Hindi language capabilities**. However, **technical limitations** (OCR errors, dashboard flaws, network dependency) and **pedagogical alignment gaps** (feedback quality, teacher override) must be addressed to realize its full potential. Prioritizing **offline support, OCR accuracy, and teacher collaboration** will be critical for scalability and adoption in diverse educational contexts.